Four-class LVEF classification from 12-lead ECG printout images

Complete record of experiments (all branches, Phases 1–7) and final result

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Abstract

Task. Classify left-ventricular ejection fraction (LVEF) into four classes (Normal, Mild, Moderate, Severe) from images of hospital 12-lead ECG printouts. Labels come from echocardiography reports (Teichholz EF) linked to the ECGs by patient-name matching.

Data and protocol. 837 development ECGs (794 patients) and a locked test set of 210 ECGs.

- Every model was evaluated with patient-grouped 5-fold cross-validation repeated 3 times, with all choices nested inside the training folds.
- Comparisons use corrected repeated-CV tests with Holm correction.
- Each study was pre-registered before it was run.

What was tested. Across seven phases and the branches of the repository:

- image CNNs, foundation-model embeddings (ECGFounder, ECG-FM) and their fine-tuning;
- LSTM, CNN-LSTM and 1-D ResNet models, MiniRocket, median-beat and clinician features;
- ensembles, stacking and two-stage models;
- label-cleaning schemes, threshold and band sweeps, and an external model trained on about 100,000 patients.

Four-class result. Four-class accuracy on all patients converges at 0.55–0.57 (best image-only 0.571 [0.542, 0.600]; locked test 0.553 [0.485, 0.621]). Mild is the hardest class.

Final model: image-only triage. It answers only when it is confident and refers the rest to echo.

- **Cross-validation:** 56 % of patients answered, with **0.694** [0.655, 0.736] accuracy.
- **Locked test, scored once:** 50 % answered, with **0.747** [0.660, 0.837] accuracy.
- 85–90 % of its confident "Normal" answers exclude moderate or severe dysfunction.

The limit. The ECG estimates EF to about ±13 points against class bands 10–12 points wide. Improvement needs ID-linked Simpson EF labels, digital ECGs, more Mild and Moderate patients, or large-scale public pre-training, which is under way.

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1 Objective

Estimate the LVEF class of a patient from the image of a standard hospital 12-lead ECG printout:

- **Normal:** EF $> 52\,\%$ (men) or $\geq 54\,\%$ (women);
- **Mild:** $41\,\%$ up to the Normal limit;
- **Moderate:** 31–$<41\,\%$;
- **Severe:** $< 31\,\%$.

The owner's requirements, as they evolved over the project:

1. four-class only (no binary simplification), with the original labels as the primary result;
2. image only: no printed machine measurements, age or sex as model inputs;
3. every class detected, with Mild recall $\geq 15\,\%$;
4. the highest accuracy obtainable without manipulating labels or the evaluation set.

2 Data

- **ECGs:** 1,047 single-centre 12-lead ECG PDFs from 993 patients (2020–2024: 232 / 294 / 226 / 295), in the standard 3×4 + rhythm layout. Each lead is 2.5 s long; lead II also appears as a 10-s rhythm strip. Class totals: Normal 480, Mild 232, Moderate 221, Severe 114.
- **Waveforms:** the traces are vector drawings and are digitised exactly from the PDF.
- **Split:**
  – a **development set** of 837 ECGs from 794 patients, used for all modelling;
  – a **locked test set** of 210 ECGs from 199 patients, with no patient overlap.
  The original project and Phase 1 scored models on the test split. From Phase 2 on it was locked: never used for training or selection, and scored once at the end with the frozen model (Section 5).
- **Labels:** Teichholz EF from echocardiography reports, linked to each ECG by fuzzy patient-name similarity. The yearly ECG–echo sheets (2020, 2021, 2022, 2024) contain no patient identifier.
- **Development class counts:** Normal 384, Mild 185, Moderate 177, Severe 91.

Data-quality audit (Phases 4–6B).

- All 837 development ECGs link to at least one echo.
- The folder label agrees with the nearest echo for 93.4 % of ECGs (55 disagree).
- Link quality: 391 links have a name-match score < 100; 192 have the nearest echo more than 30 days away.
- Echo consistency: 85 ECGs have a linked echo of another class, and 77 have an EF spread of more than 10 points.
- 221 ECGs lie within 3 EF points of a class limit.
- The 2021 sheet's ECG dates were day/month-swapped; this was corrected (C6A-1) and does not change labels.
- 29 near-identical name pairs exist under different patient keys: 15 within development, 11 straddling development and test (excluded from the locked-test primary analysis), and 3 within test.

3 Evaluation protocol

- **Cross-validation:** patient-grouped 5-fold CV $\times$ 3 repeats, identical folds for every model (`phase_2/configs/folds_D.csv`).
- **Nested choices:** hyper-parameters, thresholds and decision rules are chosen only inside the training folds (inner CV).
- **Confidence intervals:** corrected repeated-CV 95 % CIs (Nadeau–Bengio, $n_{\text{train}} = 535$, $n_{\text{test}} = 167$, 15 cells).

- **Comparisons:** paired corrected t-tests with Holm correction. A *genuine improvement* requires Holm $p < 0.05$, the same sign in all 3 repeats, and admissibility.
- **Admissibility:** every class recall ≥ 0.15. Eligibility: 3 repeats $\times$ 3 seeds.
- **Pre-registration:** plans and dated amendments were committed before running, so results cannot steer the design.

4 Summary of all work

Table 1: Every phase at a glance. Accuracy is four-class: on the test split for the original project and Phase 1, and in development cross-validation for Phases 2–7. Detail is in Sections 6–13.

Phase	What was tried	Best result	Conclusion
Original	ECGNet, ResNet-18, EfficientNet, MobileNet, ConvNeXt, HOG+SVM, 1-D ResNet, header LR, XGBoost, fusions	0.552 (header + ResNet-18; test)	the header numbers carried most of the signal; Mild recall 11 %
Phase 1	ECGFounder frozen / fine-tuned, ordinal heads, fusions	0.524, macro-F1 0.448 (A2; test)	foundation model best balanced; test set exposed
Phase 2	audit, pre-registration, development CV, input variants	0.544 (header + ECGFounder MLP)	rigorous protocol fixed
Phase 3	clinical features, gated / attention fusion, ensembles, calibration	0.547 (C1 vote)	plateau at 0.53–0.55
Phase 4	EF regression, multitask, label-quality training	0.560 (n.s.)	information limit: $r = 0.66$ vs 0.80 needed
Phase 5	engineered features, LightGBM, stacking, hierarchy, decision rules, augmentation	**0.561** (C3, selected)	0 of 43 genuine improvements
Phase 6	fine-tuning, ECG-FM, inputs, digitisation	0.561 (unchanged)	no candidate passed the screen
Phase 6A/6B	Normal 270, borderline relabelling, T50/T55, Normal sizes, LSTM, CNN-LSTM, fusions	0.569 (T55, n.s.)	LSTMs significantly worse; definition effects only
Label studies	VER, BRES, CLEAR, SUB591, HC30, MMCLEAN, NF, MILDFIX	≤ 0.565 on all ECGs	label noise explains < 1 point
Phase 7	multi-scale, soft labels, contrastive, 1-D ResNet, learning curve, cascades	0.568 (cascade, not admissible)	**triage 0.694 on 56 %**
Image-only push	two-stage, 10-s reconstruction, MiniRocket, median beats, stacking, EchoNext-Mini	**0.571** (ULT-stack)	plateau 0.57; external model adds nothing
Band sweep	3,313 class definitions	0.626 (ESC 2021)	sensitivity analysis only
Locked test	frozen image-only triage model, scored once	**0.747 on 50 %**; 4-class 0.553	development estimates confirmed

5 Final result: the image-only triage model (the "69 %" model)

5.1 Why a triage model

No model of any family exceeded about 0.57 four-class accuracy on *all* patients (Sections 7–13). Errors concentrate on ECGs whose echo EF lies near a class limit (Figure 8). A model that *answers only when it is confident* and refers the rest to echocardiography therefore turns the same information into a clinically usable result. The owner chose this framing ("triage + Mild limitation"). The model was frozen before it was tested.

5.2 Model definition (frozen 2026-11-06, SHA-256 `09808d12...`)

- **Input:** the ECG drawing only.
  - ECGFounder embedding of the waveform digitised from the PDF;

– 118 measurements made on the drawing (clinician-style features);
– no printed machine text, no age.
The patient's sex is used only to place the Normal/Mild cut (52 % men, 54 % women), exactly as in the label rule.

- **EF estimate:** the average of a ridge and a LightGBM EF regressor (unverified echo links weighted 0.3).
- **EF grading:** Normal above the sex limit; Mild 45.5 up to the limit; Moderate 35–45.5; Severe < 35.
- **Answer rule:** the model answers only when (a) the grade agrees with the image-only four-class classifier IMG-C3 (prior rule $\alpha = 0.5$), and (b) the EF estimate is at least 3.05 EF points from every cut. Otherwise it outputs "refer to echo".
- **How the values were chosen:** every threshold was set inside the training folds (nested). The frozen copy was refitted on all 837 development ECGs and checked for exact reproduction.

5.3 Results

Table 2: The final image-only model in cross-validation and on the locked test set (scored once, owner-authorised, 2026-11-06). 95 % CIs: corrected repeated-CV (development) and patient-cluster bootstrap (test).

	Cross-validation (837 ECGs)	**Locked test** (199 ECGs)
Triage: share of patients answered	56 % [48, 63]	49.7 % [42.9, 56.7]
Triage: accuracy on answered patients	**0.694** [0.655, 0.736]	**0.747** [0.660, 0.837]
Confident "Normal" answers with EF ≥ 41 %	90 % [86, 94]	85 % [77, 92]
EF < 41 % patients given a confident "Normal"	15 % [8, 22]	21 % [11, 31]
Four-class accuracy, all patients (IMG-C3)	0.558 [0.504, 0.611]	0.553 [0.485, 0.621]
Macro-F1, all patients	0.441	0.453 [0.375, 0.525]
Mild precision / recall (all patients)	0.29 / 0.19	0.32 / 0.32

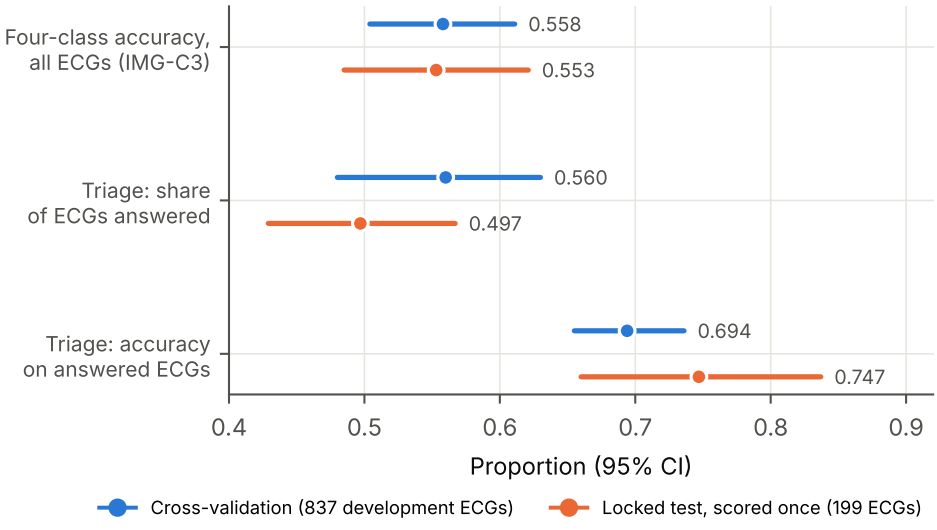


Figure 1: Development estimates carried over to unseen patients: every cross-validation value lies inside its locked-test confidence interval.

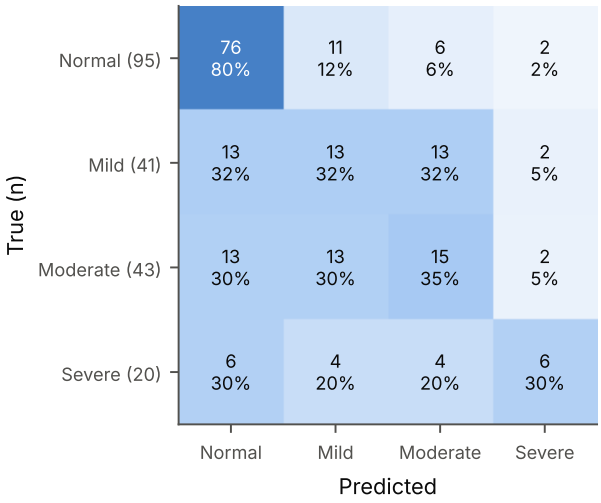


Figure 2: Locked-test confusion matrix of the four-class IMG-C3 output (counts and row percentages). Normal is recognised well; Mild and Moderate are split evenly across Normal, Mild and Moderate.

Locked-test details.

- 199 of the 210 locked ECGs form the primary set. 11 were excluded before scoring because their patient key forms a near-identical name pair with a development patient.
- The 199 ECGs come from 186 patient clusters.
- **Caveat.** The same 210 ECGs had been scored by the original project and in Phase 1 (about 40 models). The frozen triage model was developed in Phases 2–7 without any access to them, so the test is independent of its development. It is still not a pristine first look at these patients, and a new prospective test set is the stronger confirmation.
- Triage calls on the locked test: 88 Normal, 10 Moderate, 1 Severe, and no Mild.
- Sensitivity analysis on all 210 ECGs: four-class accuracy 0.538; triage 48 % answered at 0.743.

5.4 How to state the result

> From the ECG image alone, the model gives a confident answer for about half of the patients (56 % in cross-validation, 50 % on the locked test), with 0.69–0.75 accuracy on those answers. Most confident answers are "Normal", and 85–90 % of them exclude moderate or severe LV dysfunction. The remaining patients are referred to echocardiography. Four-class accuracy on all patients is about 0.55–0.57. Mildly reduced EF cannot be identified reliably from these printouts.

Do not claim:

- "70 % accuracy" without saying *on answered patients* and giving the answered share;
- that the triage detects Mild (it made no Mild calls on the locked test);
- that it is a stand-alone rule-out: 15–21 % of patients with EF < 41 % received a confident "Normal".

The 210 locked ECGs have now been used, so any further claim needs new ECGs (see the test kit, Appendix C).

6 Original project and Phases 1–5

6.1 Original project (baseline)

- **Goal:** a CNN on the cropped, grid-filtered ECG page image, predicting the four-class LVEF category and a heart-failure yes/no screen.
- **Location:** code on `main` (`ecg_ef/`); results and write-up on `claude/funny-gauss-udlmv0`.
- **Evaluation:** scored on the fixed 210-ECG test set.

Table 3: Original project, fixed test set ($n = 210$).

Model	Accuracy	Bal. acc.	Macro-F1	QWK	HF AUC
HOG + SVM	0.481	0.322	0.283	0.229	0.737
ECGNet (custom CNN)	0.471	0.360	0.262	0.389	0.745
ResNet-18 (selected image model)	0.486	0.356	0.323	0.290	0.735
EfficientNet-B0 / MobileNetV3-L	0.300 / 0.143	0.318 / 0.243	0.236 / 0.111	–	0.712 / 0.649
ConvNeXt-Atto (two learning rates)	0.395 / 0.443	0.283 / 0.289	0.262 / 0.262	–	0.618 / 0.586
1-D ResNet ×5 (digitised waveform)	0.457	0.364	0.358	0.410	0.752
Header logistic regression	0.514	0.442	0.421	0.422	0.795
XGBoost on engineered features	0.490	–	0.373	0.382	0.742
Header 0.6 + ResNet-18 0.4 (default)	**0.552**	0.454	0.433	0.447	0.791
Always “Normal”	0.457	0.250	0.157	0	–

Findings:

- Mild and Moderate were mostly called Normal: Mild recall was 6 % for ResNet-18 and 11 % for the default.
- HOG + SVM matched the CNNs, and large CNNs trained from scratch failed.
- The printed header numbers gave the largest gain.

6.2 Phase 1: foundation model, ordinal heads (fixed split, test scored)

Goal: ECGFounder frozen (A1) or fine-tuned (A2), ordinal heads (B) and EF regression (C). Branch `research/ecg-ef-model-improvements`.

Table 4: Phase 1, fixed test set.

Model	Accuracy	Bal. acc.	Macro-F1	Mild recall	HF AUC
Default (header + ResNet-18), reproduced	0.552	0.454	0.433	0.106	0.791
A1 frozen ECGFounder	0.500	0.421	0.412	0.234	0.810
A2 fine-tuned ECGFounder, 3-seed ensemble	0.524	**0.464**	**0.448**	**0.255**	0.812
A2 ordinal (cumulative link)	0.510	0.417	0.393	0.170	0.790
A1 CORAL ordinal	0.390	0.376	0.345	0.489	–
Header + A2 fusion	0.505	0.445	0.425	0.213	0.819

Findings:

- Ordinal heads lowered accuracy.
- EF regression could not be run: no EF values were available then.
- A claimed significance ($p < 0.001$) was retracted in an erratum; the corrected p is 0.12.
- About 40 models were scored on the test set, so it became *exposed*. From Phase 2 on, all modelling used development CV only.

6.3 Phase 2: audit, pre-registration, development CV

Phase 2 fixed the statistics, defined the 837-ECG development set and its fixed folds, and re-ran all models in cross-validation. The test set was not scored.

Table 5: Phase 2, development CV (mean of 3 repeats unless marked r0 = repeat 0).

Model	Accuracy	Bal. acc.	Macro-F1	Mild recall	S3
Header, numbers only (LR)	0.489	0.414	0.393	0.162	0.432
Header, text only (r0)	0.415	0.351	0.338	0.151	0.368
ECGFounder linear / + distance penalty	0.528 / 0.530	0.442 / 0.449	0.434 / 0.444	0.214 / 0.247	0.468 / 0.474
ECGFounder MLP (3 seeds)	0.542	0.457	0.446	0.207	0.481
Header + ECGFounder MLP fusion	**0.544**	0.466	0.440	0.141	**0.483**
ECGFounder fine-tuned (r0)	0.530	0.445	0.432	0.178	0.469
ECGFounder CORAL / cumulative link (r0)	0.421 / 0.474	0.385 / 0.412	0.385 / 0.409	0.411 / 0.249	–

Input variants (S3):

- aligned 0.470;
- tiled 0.432;
- lead II only 0.407.

6.4 Phase 3: towards 60–65 % (development CV)

Table 6: Phase 3, development CV.

Model	Accuracy	Bal. acc.	Macro-F1	Mild F1	S3
Header + clinical LR	0.501	0.440	0.424	0.269	0.455
3-way MLP (+ clinical)	0.541	0.466	0.442	0.202	0.483
Gated / attention fusion	0.515–0.533	–	–	–	0.458–0.473
Vote ensemble (= C1)	**0.547**	**0.467**	**0.455**	0.255	**0.490**
Stacked LR	0.547	0.473	0.442	–	0.487
MLP fusion + nested prior calibration (= C2)	0.568	0.463	0.442	0.159	–
Per-class thresholds	0.571	0.443	0.399	0.031	–

Findings:

- More than 20 model families plateau at 0.53–0.55.
- Mild one-vs-rest AUROC is 0.53–0.55 in every model.
- The calibration "gain" comes from predicting Normal more often (C2's Mild recall is 0.11, so it is not admissible).

6.5 Phase 4: EF regression, multitask learning, label quality

Table 7: Phase 4, development CV.

Model	Accuracy	Bal. acc.	Macro-F1	Mild F1	S3
Vote ensemble (benchmark)	0.547	0.467	0.455	0.255	0.490
Train without disagreeing labels	0.554	0.471	0.449	0.197	0.491
Multitask (8 variants)	0.545–0.551	–	–	0.189–0.214	0.484–0.490
Vote + MLP-regression member	0.560	0.458	0.446	0.226	0.488
EF regression, tuned cut-points	0.546	0.435	0.432	0.248	0.471
EF regression, clinical thresholds	0.558	0.416	0.418	0.287	0.464

EF regression quality:

- MAE 10.4 and R^2 0.43 (training-mean baseline MAE 15.6).
- $r = 0.66$ overall and 0.74–0.75 on the high-confidence cohort.
- AUROC 0.82 for EF $< 41\,\%$.

Conclusions:

- 0 of 26 confirmatory models improved on the vote.
- 60 % accuracy needs $r \approx 0.80$, against the observed 0.66: the limit is information, not the model.

6.6 Phase 5: features, boosting, stacking, decision rules (selection of C3)

Table 8: Phase 5, development CV (min recall = lowest class recall; admissible if ≥ 0.15).

Experiment	Accuracy	Min recall	Verdict
E1 MLP + all engineered feature families	0.554	0.164	no gain (p 0.59)
E2 LightGBM	0.544	0.184	no gain
E3 stacking (weighted / unweighted)	0.547 / 0.567	0.184 / 0.041	no gain / not admissible
E4 hierarchical model	0.567	0.067	not admissible
E5 vote + nested prior rule (= C3)	**0.561** [0.520, 0.603]	0.177	**selected**
E6 signal-quality gating	0.546	0.164	no gain
E7 mixup / augmentation	0.552 / 0.533	–	n.s.
E7 unweighted cross-entropy	0.572	0.144	not admissible
Original default re-run in CV (r0): header + ResNet-18 / ResNet-18	0.495 / 0.471	–	reference

Conclusions:

- 0 of 43 candidates is a genuine improvement.
- 65 % lies outside every model's CI.
- On the 50 % most confident ECGs, accuracy is 0.66–0.68. This is the first sign of the triage result in Section 5.

7 Phase 6: advanced model improvement

Goal. Raise the information extracted per ECG, using foundation models and better inputs, in a two-stage pre-registered design. Stage 1 is a cheap screen; Stage 2 is the full run only for candidates that pass.

- **Screen criterion:** EF correlation $r \geq 0.69$, or macro-F1 ≥ 0.465 with all recalls ≥ 0.15.
- **Branch:** `research/phase6`. **Report:** `phase_6/reports/PHASE6_FINAL_REPORT.md`.

Table 9: Phase 6 experiments (development CV).

Experiment	Result	Reason
A1/A2 ECGFounder fine-tuning (stage 6; stages 5–6)	EF r 0.650 vs 0.652 frozen	early stopping at epochs 4–15; about 535 training ECGs per fold overfit the backbone
B ECG-FM (MIT); ECGFounder + ECG-FM	r 0.609 / 0.636 vs 0.638	a second foundation encoder carries the same or less EF information
HuBERT-ECG, ST-MEM	not run	non-commercial licences, owner decision pending
ECG-JEPA, MERL	not run	weights hosted on Google Drive, blocked by network policy
C extra ECGs	not run	no extra PDFs or waveforms exist
D1 digitisation quality	no effect	EF error unrelated to quality flags ($\rho = -0.04$)
D2 10-s lead-II strip	no added information	r 0.647 without vs 0.638 with

Outcome:

- No candidate passed the Stage-1 screen, so C3 stayed the selected model (0.561 [0.520, 0.603]).
- The best representation remains the frozen ECGFounder embedding ($r \approx 0.66$).
- Merging the 15 within-development near-identical name pairs changed accuracy by -0.002 to -0.010.

8 Phase 6A: Normal/Mild boundary and LSTM models

Goal (faculty guide's brief). Test three label and training variants, and whether a recurrent model reading the waveform beats the existing models:

- **A270:** 270 Normal training ECGs;
- **B2:** borderline Mild with EF ≥ 50 relabelled Normal;
- **B3:** uncertain borderline ECGs removed from training;
- **LSTMs:** variants L0–L4 with nested selection (LSTM-sel), a fixed variant, and a CNN-LSTM.

Figure 3 shows the printout layout and the LSTM family. Branch `research/phase6`, folder `phase_6a_normal_mild_lstm`.

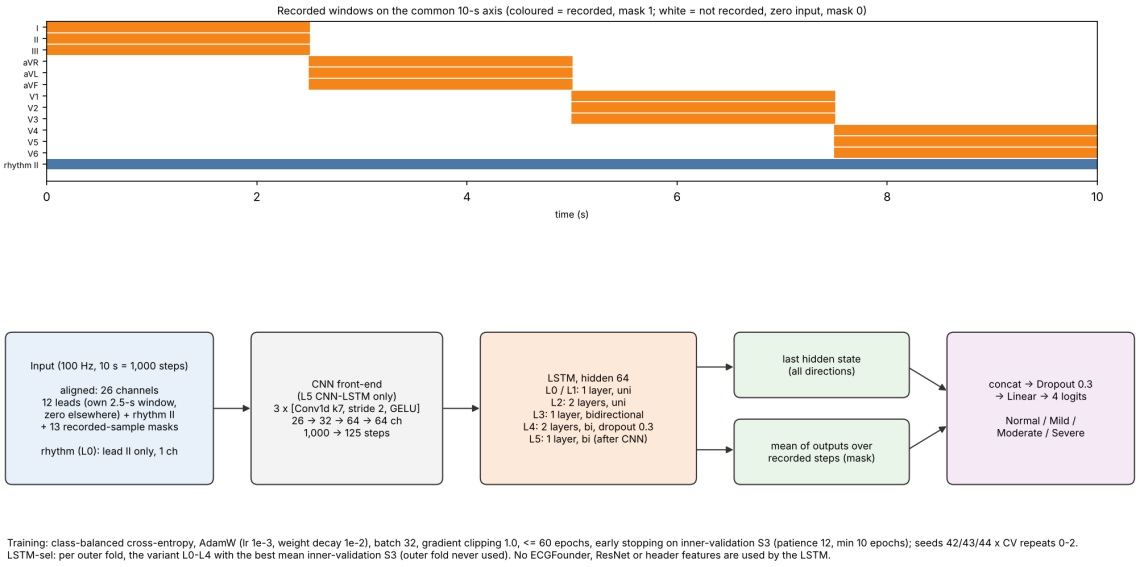


Figure 3: Top: the printout gives each lead only its own 2.5-s column; only lead II covers 10 s. Bottom: the LSTM / CNN-LSTM family tested in Phases 6A–6B.

Table 10: Phase 6A key results (original labels; corrected 95 % CI). "Adm." means every class recall ≥ 0.15.

Model	Adm.	Accuracy [95 % CI]	Macro-F1	Mild F1	Mild recall
C3 Phase 5 selected	yes	0.561 [0.520, 0.603]	0.448	0.221	0.177
C1 vote ensemble	yes	0.547 [0.504, 0.590]	0.455	0.255	0.220
C2 calibrated MLP	no	0.568 [0.534, 0.603]	0.442	0.160	0.112
ECGFounder fine-tuned (A1)	yes	0.536 [0.492, 0.581]	0.436	0.202	0.164
ECGFounder frozen, linear	yes	0.528 [0.484, 0.573]	0.434	0.244	0.214
C3 recipe [A270]	yes	0.561 [0.518, 0.604]	0.446	0.237	0.193
C3 recipe [B2]	no	0.554 [0.511, 0.596]	0.434	0.187	0.142
C3 recipe [B3]	no	0.559 [0.518, 0.600]	0.426	0.144	0.103
LSTM L0 (rhythm only)	no	0.426 [0.387, 0.466]	0.257	0.065	0.040
LSTM L2 (best single)	no	0.483 [0.437, 0.529]	0.352	0.114	0.076
LSTM-sel (nested)	no	0.470 [0.424, 0.515]	0.344	0.122	0.081
CNN-LSTM L5	yes	0.484 [0.435, 0.532]	0.368	0.219	0.191

Findings:

- Mild is the main difficulty. Of 185 Mild ECGs, C3 predicts 79 as Normal, 52 as Moderate and 21 as Severe, and only 33 as Mild.

- A270, B2 and B3 give no improvement (all Holm $p = 1.00$). The apparent B2 gain on B2 labels is a *definition effect*, not a training effect.
- The only significant results are the *inferiority* of the standalone LSTMs (about -0.09 accuracy vs C3).

9 Phase 6B: EF thresholds, Normal training size and the LSTM family

Goal.

- Clinically motivated threshold variants: T50 (Normal $\geq 50\,\%$) and T55 (Normal $\geq 55\,\%$).
- Normal training sizes 300 / 270 / 240 / 200.
- Removal of uncertain cases (XUNC).
- CNN-LSTM with and without the header, and early or late fusion with ECGFounder.

Branch `research/phase6`, folder `phase_6b_threshold_sampling_lstm`.

Table 11: Phase 6B: existing recipe under each configuration, scored on the *original* labels (accuracy / Mild recall).

Config	MLP fusion	C1 vote	C3	LSTM-fixed
ORIG	0.544 / 0.141	0.548 / 0.222	0.562 / 0.178	0.467 / 0.067
T50	0.548 / 0.148	0.547 / 0.205	0.554 / 0.142	0.466 / 0.068
T55	0.544 / 0.166	0.544 / 0.207	**0.569** / 0.186	0.459 / 0.097
XUNC	0.545 / 0.124	0.542 / 0.169	0.559 / 0.103	0.462 / 0.077
N300	0.546 / 0.142	0.543 / 0.220	0.559 / 0.184	0.464 / 0.068
N270	0.545 / 0.142	0.547 / 0.225	0.558 / 0.195	0.458 / 0.086
N240	0.539 / 0.155	0.545 / 0.227	0.558 / 0.209	0.474 / 0.072
N200	0.541 / 0.150	0.546 / 0.234	0.553 / 0.220	0.470 / 0.067

Table 12: Phase 6B LSTM family and fusions (original labels).

Model	Accuracy [95 % CI]	Macro-F1	Mild recall	Parameters	Train min/run
F4 vote + LSTM-sel + prior rule	0.559 [0.517, 0.600]	0.449	0.182	C1 + 24k–147k	3.52
F3 vote + LSTM-sel	0.553 [0.508, 0.597]	0.453	0.198	C1 + 24k–147k	2.29
F2 CNN-LSTM + header + f6	0.551 [0.517, 0.585]	0.452	0.166	198,820	0.26
F1 LSTM-fixed + header + f6	0.540 [0.507, 0.573]	0.439	0.151	130,372	0.29
L6 CNN-LSTM + header	0.497 [0.456, 0.539]	0.389	0.182	116,644	0.27
L5 CNN-LSTM	0.484 [0.435, 0.532]	0.368	0.191	116,580	0.38
LSTM-fixed L3	0.467 [0.417, 0.516]	0.337	0.067	48,132	0.44

Findings:

- No configuration or fusion improves on C3; all Holm $p = 1.00$. T55 training gives the highest admissible figure on original labels (0.569), but $+0.007$ is not significant.
- Reducing Normal training ECGs shifts the prior (Mild recall rises, Normal recall falls) without better discrimination (Figure 4).
- Under T50 the higher scores are a definition effect: 27 hard borderline Mild ECGs become Normal.

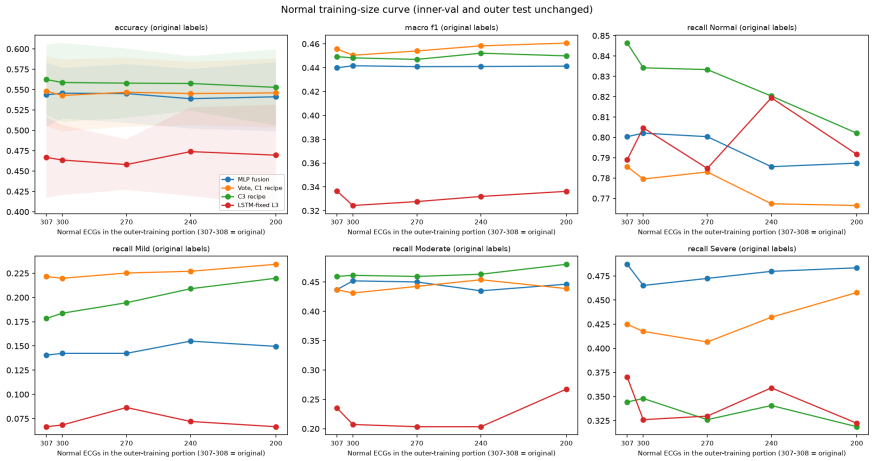


Figure 4: Normal training size vs accuracy and recall (Phase 6B).

10 Label-quality and cohort studies (exploratory)

These studies answered the owner's questions about hard borderline cases, label cleaning and "reliable" subsets. All are on branch `research/phase6` (`phase_6b_threshold_sampling_lstm/`), each with a dated amendment written before it was run.

Table 13: Label / cohort studies. Accuracy on all 837 ECGs unless stated.

Study	Result	Conclusion
VER: 15 labels corrected to a high-confidence echo	C3 0.565 (+0.004)	label noise explains <1 point
VERB / BRES: resolve or remove borderline cases	gains only by abandoning Mild	not admissible
CLEAR: train without border cases	0.548 all; 0.668 on clear-cut; Mild recall 0.13	subset gain, not a model gain
SUB591: N 200 / Mi 150 / Mo 150 / S all	0.548 all; 0.489 on subset	no gain
HC30: train and test on 395 exact-name, $\leq$30-day links	0.594 (C3 trained on all: 0.598)	reliable labels alone do not help
HC30F: + all ECG factor families	0.603 on HC30 (Mild recall 0.25)	+0.005, noise level
MMCLEAN: remove suspect Mild/Moderate (217)	0.547 all; 0.677 on kept 620 (Mild recall 0.09)	higher numbers come from easier subsets
NF / NFS: noise-free training	0.546 / 0.552 all; C3 0.707 on 264 clean, clear-cut	cleaning does not raise all-ECG accuracy
MILDFIX F1–F3: Mild-aware, ordinal, EF-based rules	0.537–0.560	trade accuracy for Mild
Circularity check: relabel border cases to model output	0.764, but random guessing also rises 0.244 $\rightarrow$ 0.470	**invalid** (circular); excluded

Selective prediction (answer only above a nested confidence threshold) first reached about 0.71 on 22 % of ECGs (MMCLEAN, Figure 5). It was refined in Phase 7 into the final triage model (Section 5).

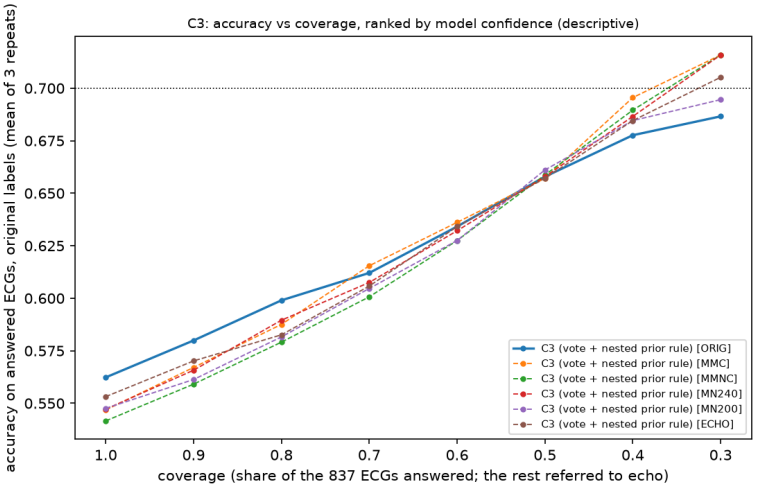


Figure 5: Accuracy vs share of ECGs answered (selective prediction).

11 Phase 7: representation and head experiments, learning curve, triage

Branches: `research/phase6` (`phase_7/`) and `research/phase7-testkit` (frozen model and locked test). **Pilot gate:** accuracy $\geq +0.010$ or Mild F1 $\geq +0.03$, with macro-F1 ≥ -0.010. No candidate passed (Table 14).

Table 14: Phase 7 pilots (seed 42, 3 repeats).

Exp.	Candidate	Accuracy	Macro-F1	Mild P / R / F1
ref	MLP fusion [header ‖ f6]	0.544	0.436	0.283 / 0.132 / 0.179
E1	stage-5 mean ‖ max ‖ std	0.517	0.409	0.263 / 0.146 / 0.187
E1	per-printed-column stage-5 mean	0.527	0.431	0.271 / 0.157 / 0.199
E1	stage-4 mean ‖ f6 (multi-scale)	0.529	0.425	0.250 / 0.141 / 0.178
E1	150 ECGFounder diagnostic outputs	0.531	0.412	0.287 / 0.117 / 0.165
E2	multitask CE + verified-EF regression	0.545	0.440	0.289 / 0.142 / 0.191
E2	boundary-aware soft labels	0.548	0.433	0.303 / 0.123 / 0.174
E2	EF-weighted supervised contrastive	0.546	0.436	0.276 / 0.130 / 0.176
E3	multi-lead 1-D ResNet (repeat 0)	0.496	0.384	0.264 / 0.178 / 0.213
E3	1-D ResNet + header (repeat 0)	0.485	0.378	0.243 / 0.189 / 0.213

E5: learning curve (Figure 6):

- each doubling of training data gives +1.4 accuracy points, with no trend in Mild F1;
- the EF error SD falls by 0.44 points per doubling (0.66 on verified-EF ECGs);
- extrapolation: about 10^5 ECGs of current label quality would give about 0.65.

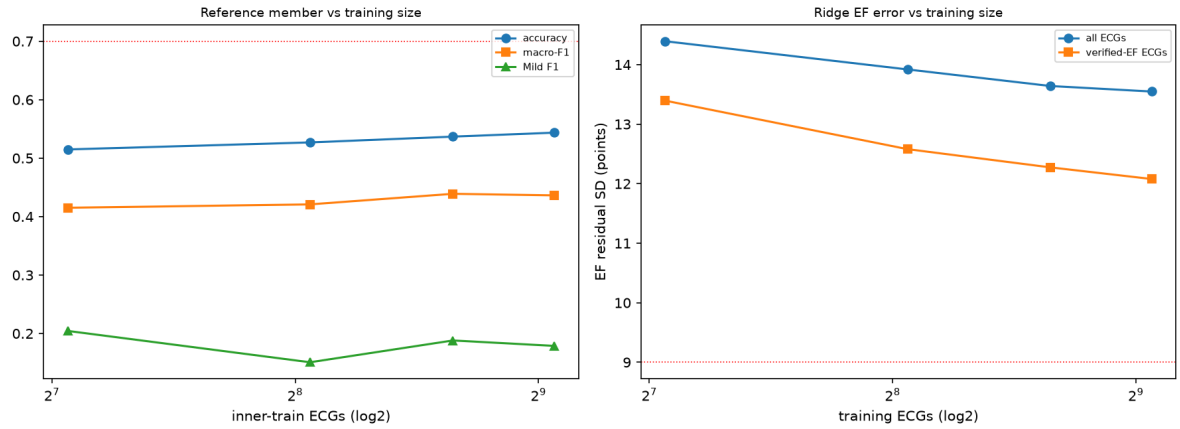


Figure 6: Phase 7 learning curve: classification metrics (left) and EF error (right) vs training size.

E6: header-free cascades.

- EF-grading stage 1 alone: 0.693 on 53 % answered (G2T-img); 0.694 on 56 % when it must agree with IMG-C3 (AGT-img, the final triage).
- As complete four-class classifiers the cascades reach 0.548–0.568, no genuine gain.
- Stage 1 answers almost only Normal.

E7: Mild-aware selective prediction.

- Accuracy ≥ 0.70 together with detecting ≥ 10 % of Mild patients is *infeasible* in all 15 training folds.
- Even the most confident Mild calls are right only 20–50 % of the time.

12 Final push on image-only four-class accuracy

At the owner's request (image only, Mild ≥ 15 %), four further pre-registered studies were run. They are on branch `research/phase6`: `HIER_REPORT.md`, `ENH_REPORT.md`, `ULT_REPORT.md` and `ECHONEXT_REPORT.md`.

Table 15: Image-only four-class models, all 837 ECGs (no header, age or sex input).

Model	Accuracy [95 % CI]	Recall N / Mi / Mo / S
HIER: two-stage (N / middle / S, then Mi vs Mo)	0.558 [0.522, 0.594]	0.86 / 0.17 / 0.41 / 0.34
HIER-ext (wider tuning grid)	0.570 [0.527, 0.612]	0.88 / 0.16 / 0.43 / 0.34
ENH: synth 10-s reconstruction → ECG-Founder (M2)	0.528	0.76 / 0.18 / 0.44 / 0.41
ENH: same head on aligned input (D1 reference)	0.537	0.78 / 0.21 / 0.41 / 0.42
ENH: median-beat shape (M3)	0.467	0.68 / 0.19 / 0.36 / 0.33
ENH: MiniRocket (M4)	0.504	0.74 / 0.22 / 0.38 / 0.33
ENH: EF-ordinal ridge (M6)	0.534	0.95 / 0.29 / 0.00 / 0.29
ENH-vote (6 members, tuned class weights)	0.569 [0.527, 0.610]	0.88 / 0.17 / 0.46 / 0.29
ULT-stack (10 members + predicted EF)	**0.571 [0.542, 0.600]**	0.91 / 0.16 / 0.42 / 0.29
ULT2-stack (+ EchoNext-Mini external model)	0.569 [0.538, 0.599]	0.90 / 0.15 / 0.44 / 0.28
EchoNext-Mini outputs + LR alone	0.486 [0.436, 0.536]	0.72 / 0.11 / 0.36 / 0.53

Findings:

- **Two-stage:** stage 1 (Normal / middle / Severe) is right 66–67 % of the time. Stage 2 (Mild vs Moderate) is right only 53–58 %, close to a coin flip.
- **Reconstruction:** a faithful beat-synchronous 10-s 12-lead reconstruction (template $r = 0.98$) adds no EF information.
- **External knowledge:** EchoNext-Mini (Hughes et al., 2026; Poterucha et al., 2025), trained on about 100,000 US patients, transfers to the printouts (AUROC 0.79 for EF ≤ 45 %). It is weaker than the in-house EF model (0.86) and adds no accuracy.
- **Ensembles:** all ensembles plateau at 0.57 ± 0.01 (Figure 9).

13 Sensitivity to the class definition (band sweep)

- **Sweep:** 3,313 definitions (Normal limit 45–56, Mild/Moderate 35–46, Moderate/Severe 25–36, sex offset 0/2) were evaluated with one header-free EF model. 1,457 of them are feasible (every class ≥ 60 ECGs, both middle bands ≥ 5 points wide).
- **Confirmation:** the selected definitions were retrained.
- **Result:** the clinically pre-specified **ESC 2021** definition (Normal ≥ 50, Mild 41–49, Moderate 30–40, Severe < 30) gives **0.626 [0.588, 0.664]** with every class detected (Mild 17 %, Moderate 42 %).
- **Trade-off:** no definition detects both Mild and Moderate well (Figure 7).

These figures belong to a different task definition and are reported only as a sensitivity analysis.

Table 16: Confirmed band definitions (retrained header-free vote).

Definition	N / Mi / Mo / S ECGs	Accuracy [95 % CI]	κ	Detected N / Mi / Mo / S
Original limits on echo EF	415 / 170 / 165 / 87	0.594 [0.560, 0.627]	0.353	91 / 20 / 32 / 38 %
ESC 2021	446 / 139 / 176 / 76	**0.626** [0.588, 0.664]	0.384	90 / 17 / 42 / 35 %
Best Mild detection (>51 / 38 / 31)	439 / 197 / 114 / 87	0.619 [0.593, 0.645]	0.380	90 / 28 / 25 / 43 %
Best κ (>52 / 46 / 29)	428 / 70 / 272 / 67	0.634 [0.591, 0.678]	0.399	87 / 12 / 49 / 27 %

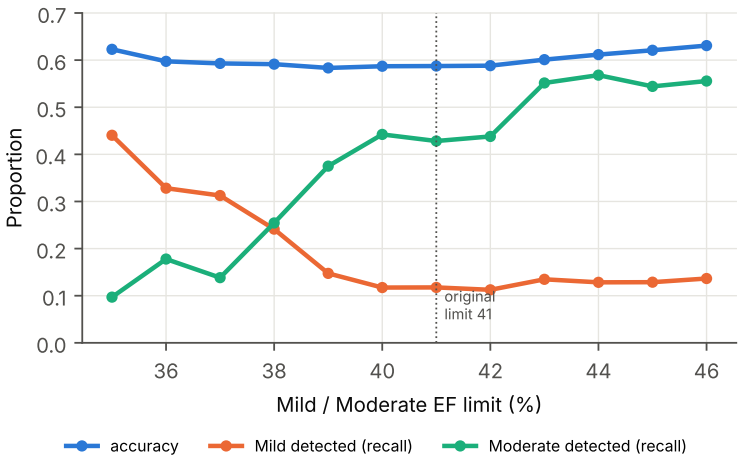


Figure 7: Moving the Mild/Moderate limit trades Mild detection against Moderate detection.

14 Why accuracy plateaus at about 0.57 on all patients

More than 15 model families, 8 label-cleaning schemes, an external model trained on about 100,000 patients, and 3,313 class definitions all land at 0.55–0.57 four-class accuracy (0.63 with the ESC

2021 limits). The convergence points to a limit in the *information* carried by these printouts and labels, not in the modelling. The evidence:

1. **EF precision vs band width.**
   - The best header-free models estimate EF with $r \approx 0.66$, residual SD about 13 points and MAE about 10.
   - The Mild band (41 to 52/54) is about 12 points wide and the Moderate band 10.
   - A simulation (true EF plus Gaussian error, cut at the class limits) gives an accuracy ceiling of 0.63 at SD 13. Reaching 0.73 needs SD 8, and 0.82 needs SD 5.
   - The models reach 0.57–0.59, close to this ceiling.
2. **Errors sit at the class limits** (Figure 8).
   - 51 % of ECGs have an echo EF within 6 points of a limit; their error rate is 62–68 %.
   - ECGs at least 10 points from every limit are classified with 79–84 % accuracy.
   - That clear-cut set contains no in-band Mild or Moderate ECG: those bands are too narrow.
3. **The ECG separates the extremes, not neighbouring grades.**
   - Pairwise AUROC: Normal vs Severe 0.90 and Normal vs Moderate 0.88, but Normal vs Mild 0.75, Mild vs Moderate 0.63 and Moderate vs Severe 0.64.
   - The best single Mild-vs-Moderate factor reaches 0.64.
   - Mild differs from Normal mainly by conduction markers: QRS width, LBBB patterns.
4. **Label uncertainty.**
   - EF comes from Teichholz M-mode reports.
   - The echo is linked to the ECG by fuzzy name matching: only 44 % of ECGs have a clean link (exact name, echo within 30 days). 391 have a name-match score < 100 and 192 an echo more than 30 days away.
   - Repeat echoes of the same ECG differ by a median of 15.4 points (an upper bound).
   - ECG–echo agreement is $r = 0.75$ for clean links vs 0.42 for echoes more than 30 days away.
   - Cleaning the labels changes all-ECG accuracy by only -0.02 to $+0.01$, because the remaining noise still exceeds the band width.
5. **Input format.**
   - The PDFs give 2.5 s per lead in four successive columns; only lead II covers 10 s.
   - Foundation models were pre-trained on simultaneous 10-s digital ECGs.
   - Reconstruction, multi-scale and per-column embeddings did not help. Digitisation quality was unrelated to error.
6. **Sample size.** 837 development ECGs, of which 185 are Mild. The learning curve gives +1.4 points per doubling, with no Mild trend.

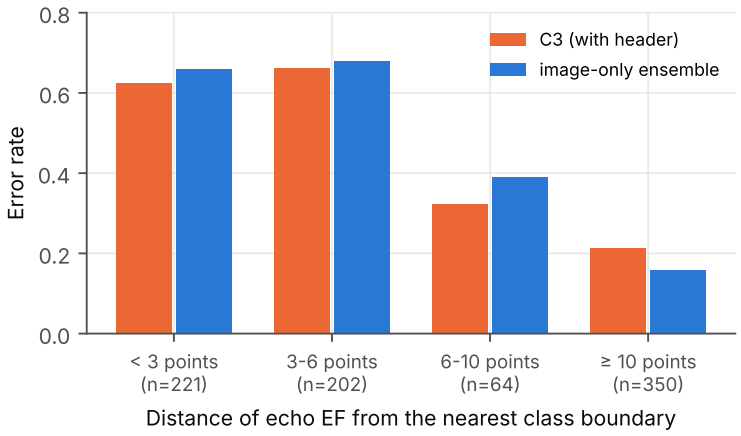


Figure 8: Error rate by distance of the echo EF from the nearest class limit (all 837 ECGs, mean of 3 repeats).

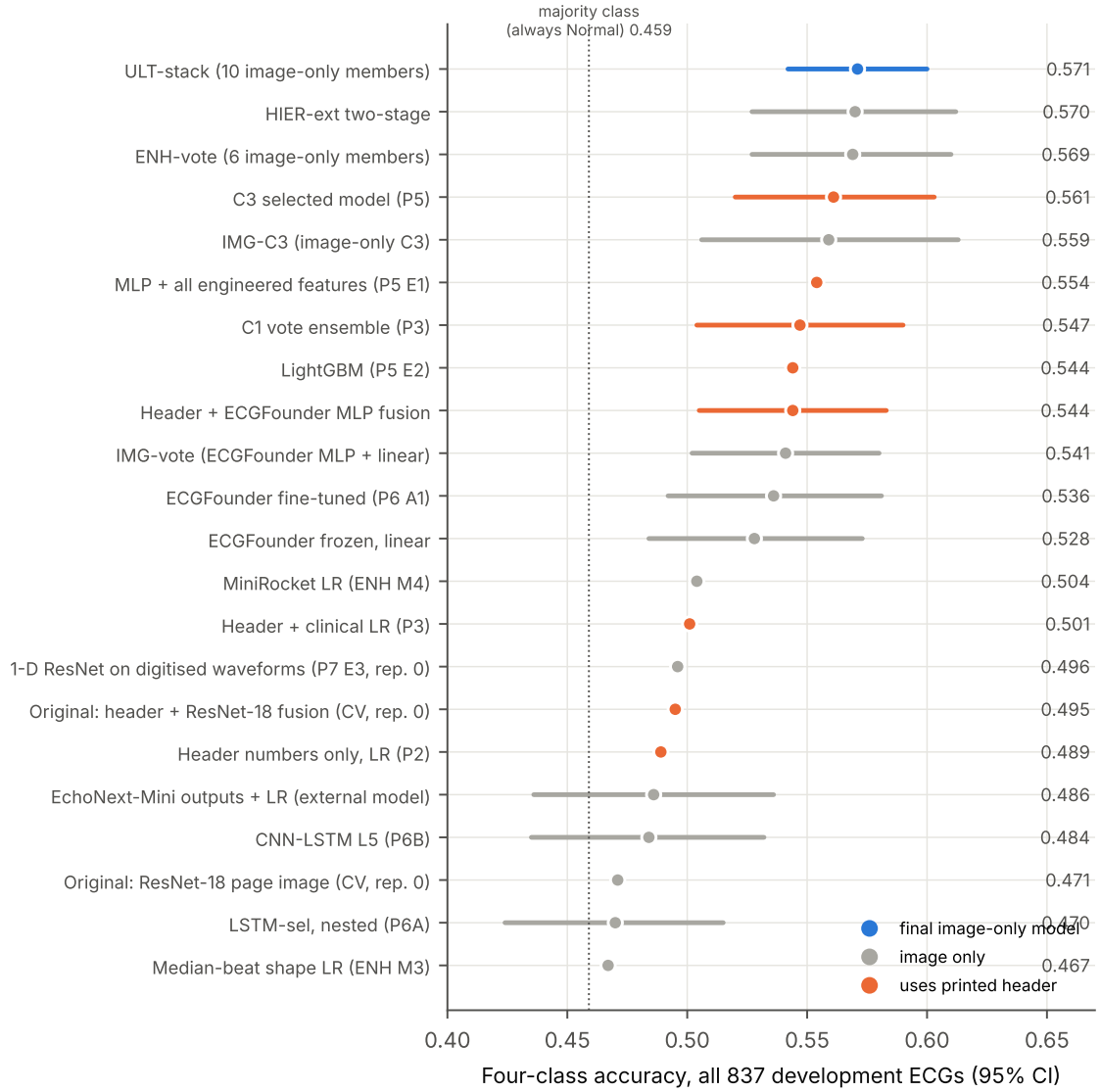


Figure 9: Four-class accuracy (all 837 development ECGs) of the main models across all phases. Image-only and header-using models converge at 0.55–0.57.

15 Recommendations and ongoing work

1. **Verified ECG–echo linkage.** Link by MRN/UHID, with the echo within days and ideally in the same admission. A prioritised re-check worklist (197 high-priority ECGs, mostly Mild and Moderate) and a fill-in template are ready. A pipeline re-trains and re-scores the model as soon as verified data return; its self-test reproduces the current result exactly (`fixes.py`).
2. **Biplane Simpson EF** instead of Teichholz, with repeat reads near the 41 % and 50–54 % limits.
3. **Digital 10-s simultaneous ECGs** (XML/DICOM exports from the ECG machine) instead of PDF printouts.
4. **More patients, enriched for Mild and Moderate.** At least 3,000 pairs, with at least 600 each of Mild and Moderate, preferably from a second centre.
5. **Public-data pre-training (in progress, branch `research/public-ef`).**
   - EchoNext (PhysioNet, credentialed) provides 100,000 ECGs with a numeric LVEF.
   - Its ECGs are cut into the hospital printout layout, labelled with the same four-class rule, used for training, and then evaluated on the 837 development ECGs with the same folds.
   - The data stay on the owner's laptop under the data-use agreement.

6. **A new prospective test set** scored once with the frozen test kit (`research/phase7-testkit`), because the locked set has now been used.

A Repository branch map

Branch	Content
`main`, `claude/funny-gauss-udlmv0`	Original project: PDF preprocessing and baseline image models (code on `main`; results and write-up on `claude/funny-gauss-udlmv0`)
`research/ecg-ef-model-improvements`	Phases 1–3: improved models, Phase 3 report, `FINAL_REPORT.md`
`research/phase4-handoff`	Phase 4: label quality, regression, multitask, final report
`research/phase5`	Phase 5: baselines, feature families, nested decision rules; selection of C3
`research/phase6`	Phase 6 (foundation-model improvement), 6A, 6B and all exploratory studies; Phase 7 (`phase_7/`); this report (`final_report/`)
`research/phase7-testkit`	Frozen header-free triage model, test kit and the one-time locked-test result
`research/public-ef`	EchoNext public-data pre-training study (code only; data stay on the laptop)

B Glossary

C1 Equal-weight vote of three members: the MLP fusion of the printed header and the ECGFounder embedding, an ECGFounder linear head (distance-penalised cross-entropy), and a header + clinical logistic regression.
C2 Calibrated MLP fusion (not admissible: Mild recall 0.11).
C3 C1 plus a class-prior decision rule tuned inside the training folds. The Phase 5 selected model.
IMG-C3 The C3 recipe without any header input (ECGFounder heads only).
Admissible Every class recall ≥ 0.15.
Definition vs training effect Under changed labels, the gain of the unchanged model (definition effect) vs the gain from retraining (training effect).
Aligned / synth input The 12 × 10-s ECGFounder input built from the printout: leads at their true 2.5-s position with zeros elsewhere (aligned), or a beat-synchronous reconstruction (synth).
Triage A model that answers only when confident and otherwise refers to echocardiography.

C Integrity and reproducibility

- **Pre-registration:** every phase and exploratory study has a plan or a dated amendment committed *before* it was run. Plan hashes are stored in `configs/PLAN.sha256`.
- **Locked test:** the 210-ECG test set was never used for training or selection. It was scored exactly once, with the owner's written authorisation, by the frozen model (2026-11-06; rules committed beforehand).
- **Privacy:** only aggregate metrics are committed. Per-ECG predictions, embeddings, model weights and the ECG–echo sheets stay in git-ignored folders, and every push is privacy-scanned.
- **Excluded as invalid:** relabelling ECGs to the model's own predictions (circular), and hand-picking evaluation cases.
- **Re-running a study:** each study's code is next to its report. For example:

```
python -m research_experiments.phase_6b_threshold_sampling_lstm.ult \
       --members --combine
python -m research_experiments.phase_7.testkit.score --help
python -m research_experiments.final_report.make_figures
```

References

- Li J, et al. An electrocardiogram foundation model built on over 10 million recordings (ECG-Founder). *NEJM AI*, 2025.
- McKeen K, et al. ECG-FM: an open electrocardiogram foundation model. *JAMIA Open*, 2025.
- Hughes JW, Jing L, Finer J, et al. EchoNext-Mini: a dataset and baseline AI model for detecting structural heart disease from electrocardiograms. *NEJM AI*, 2026. doi:10.1056/AIdbp2500516.
- Poterucha TJ, et al. Detecting structural heart disease from electrocardiograms using AI. *Nature*, 2025; 644(8075). doi:10.1038/s41586-025-09227-0.
- McDonagh TA, et al. 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure. *Eur Heart J*, 2021.
- Nadeau C, Bengio Y. Inference for the generalization error. *Machine Learning*, 2003 (corrected resampled t-test).
- Dempster A, Schmidt DF, Webb GI. MiniRocket: a very fast (almost) deterministic transform for time series classification. *KDD*, 2021.